MILITARY SPECIFICATION SHEET

ELECTRON TUBES, POWER

TYPES 6816 AND 6884

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tubes described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Tetrode, UHF, ceramic-metal

Outline: See figure 1

Mounting position: Any

Weight: 2 ounces nominal

ABSOLUTE RATINGS: F1 = 1,215 MHz

Parameter: / Unit:	Type	Ef V (see note 2)	Eb Vdc	Ec1 Vdc	Ec2 Vdc	Ib mAdc	Ic1 mAdc	Pg2 W	Pp W	Pi W	T(seal) °C (see note 3)	tk sec (min)
Maximum:												
AB Audio:	6816	6.3±10%	1,000	---	300	180	30	4.5	115	180	250	60
	6884	26.5±10%	1,000	---	300	180	30	4.5	115	180	250	60
AB SSBSC:	6816	6.3±10%	1,000	---	300	250	30	4.5	115	180	250	60
	6884	26.5±10%	1,000	---	300	250 (see note 4)	30	4.5	115	180	250	60
C Telep:	6816	6.3±10%	800	-100	300	150	30	3.0	75	120	250	60
	6884	26.5±10%	800	-100	300	150	30	3.0	75	120	250	60
C Teleg:	6816	6.3±10%	1,000	-100	300	180	30	4.5	115	180	250	60
	6884	26.5±10%	1,000	-100	300	180	30	4.5	115	180	250	60
TEST CONDITIONS:	6816	6.3	1,000	Adjust	300	---	---	---	---	---	---	60
	6884	26.5	1,000	Adjust	300	---	---	---	---	---	---	60

GENERAL:

Qualification - Required

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	SYMBOL	LIMITS		UNIT
					MINIMUM	MAXIMUM	
	Quality conformance inspection, part 1	1					
1266	Total grid current	7	Ec1/Ib'= 115 mAdc	Ic1	---	-8	µAdc
1256	Electrode current (screen)	---	Ec1/Ib = 115 mAdc	Ic2	-8.0	2.0	mAdc
1261	Electrode voltage (1) (grid)	---	Ec1/Ib = 115 mAdc	Ec1	-6	-15	Vdc
1261	Electrode voltage (2) (grid)	---	Ec1/Ib = 1.0 mAdc	Ec1	---	-48	Vdc
1231	Pulsing emission	8	etd/ik = 10a	etd	---	300	v
	Quality conformance inspection, part 2						
1301	Heater current	--- ---	Type 6816 Ef only Type 6884 Ef only	If If	1.84 0.48	2.26 0.60	A A
1266	Primary grid emission (control)	---	Eg1/g1 input = 2 W; t = 30; anode and g2 grounded	Isg1	---	-2	µAdc
1266	Primary grid emission (screen)	---	Eg2/g2 input = 4.5 W; t = 30; anode and g1 grounded	Isg2	---	-3	µAdc
1366	Interelement leakage resistance cold	6	Supply voltage = 200 Vdc	R	1.0	---	Meg
2214	Power output	5	F = 400 ±20 MHz; Ib = 180 mAdc (max); Ic1 = 30 mAdc (max); t = 120; Pd = 3.3 W (max) Type 6816 Ef = 5.7 V Type 6884 Ef = 24.0 V	Po	80	---	W (useful)
1236	Power oscillation	---	F = 15 MHz; Eb = 850 Vdc; Ec2 = 300 Vdc (max); Ib = 150 mAdc; Rg1 = 4,000 ohms (max); Ic1 = 30 mAdc; Pg2 = 4.5 W (max); t = 120	Po	80	---	W (useful)
1372	Current division (method A)	---	Eb = 350 Vdc; Ec1 = -100 Vdc; egk/ib = 0.6 a; prr = 10 to 12; tp = 4,500 to 5,000 µs	egk ic1 ic2	-3.0 0 10	+17 130 70	v ma ma
1256	Electrode current (positive grid)	---	Ec1 = +2 Vdc; anode and g2 floating; no other voltages except Ef	Ic1	6	---	mAdc
1331	Direct-interelectrode capacitance	---	Use capacitance fixture per 289-JAN	Cg1k Cg1g2 Cg2p Cg1p Cpk Cg2k	11.0 15.0 4.2 --- --- 0.20	15.0 20.0 5.2 0.065 0.013 0.45	pF pF pF pF pF pF

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	SYMBOL	LIMITS		UNIT
					MINIMUM	MAXIMUM	
	Quality conformance inspection, part 3						
---	Life test	7	Group C; Ib = 180 mAdc; F = 400 ±20 MHz; Ic1 = 30 mAdc (max); Ec1 = value obtained for Ec1 in electrode voltage (2) test per part 1 herein; Type 6816 Ef = 5.7 V Type 6884 Ef = 24.0 V	t	500	---	hrs
---	Life test end points:						
2214	Pulsing emission	---	etd/ik = 7.5 a	etd	---	400	v
1266	Primary grid emission (control)	---		Isg1	---	-4.0	µAdc
1266	Primary grid emission (screen)	---		Isg2	---	-4.0	µAdc
2214	Power output	---		Po	70	---	W (useful)
1506	Heater-cycling life test	---	Group C; accelerated heater-cycling 2-1/2 min "on", 5 min "off" Type 6816 Ef = 8.5 V Type 6884 Ef = 35.0 V	---	384	---	Cycles
---	Heater-cycling life-test end point:						
1301	Heater current	---					
	Type 6816			If	1.5	2.7	A
	Type 6884			If	0.4	0.7	A
1031	Low-frequency vibration	9	Ec2 = 250 Vdc; Ec1/Ib = 10 mAdc; Rp = 2,000 ohms; Ebb = 300 Vdc	Ep	---	300	mVac
1031	High-frequency vibration	9	No voltages	---	---	---	---
1042	Shock	9	Test condition A	---	---	---	---
---	Vibration and shock end points:						
1266	Total grid current	7	Ec/Ib = 115 mAdc	Ic1	---	-8	µAdc
1261	Electrode voltage (1) (grid)	---	Ec/Ib = 115 mAdc	Ec1	-6	-15	Vdc

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NOTES:

1. The AQL for all tests listed under quality conformance inspection, part 1, shall be 0.65, inspection level II.

2. Because the cathode is subjected to considerable back bombardment as the frequency is increased, with resultant increase in temperature, the heater voltage shall be decreased depending on operating conditions and frequency to prevent overheating the cathode and resultant short life.

3. Sufficient cooling shall be supplied to maintain anode core and seal temperatures below the maximum rating unless otherwise specified.

4. The maximum dc anode current at the peak of the current is 250 mAdc for a signal having a minimum peak-to-average power ratio of 2.0. The maximum rating for a signal having a minimum peak-to-average power ratio of less than 2.0, such as is obtained in single-tone operation, is 180 mAdc. During short periods of circuit adjustment under single-tone conditions, the average anode current may also reach the level of 250 mAdc.

5. Tube shall be tested in a grid-driven amplifier circuit. Adjust Ec1 bias supply and tune circuit for maximum useful power output. The specified driver power (Pd) output shall be measured with a Bird Thru-line Wattmeter, or equivalent. Driving power output = forward power minus reflected power. Grid voltage supply shall have an effective impedance of 500-ohms maximum.

6. Before subjection to this test, the tube shall be cooled (room ambient temperature) for 30 minutes. Using the test circuit shown on the figure of method 1366, measure the resistance, in both directions, between any two adjacent electrodes (except across the heater terminals). If the resistance value is below that required, repeat this test after a 10-minute interval. If the tube fails again, the tube shall be rejected.

7. Cooling shall be set up to maintain the tube temperature at the hottest point on the tube envelope at 225 ±25°C. (This set up should be permanent and need be verified only as often as required to maintain equipment accuracy).

8. With anode, grid 1, and grid 2 tied together, apply a pulse voltage source between anode and cathode. Adjust the pulse amplitude until the specified peak cathode current is obtained. Test shall be made at the end of 1 minute or when stability is reached, whichever occurs first. The pulse voltage shall meet the following requirements:
   (a) Pulse duration 2 µs
   (b) Pulse repetition rate 60 Hz
   (c) Duty factor 0.00012

9. This test shall be performed during the initial production and once each succeeding 12-calendar months in which there is production. A regular double sampling plan shall be used, with the first sample of three tubes with an acceptance number of zero, and a second sample of three tubes with a combined acceptance number of one. In the event of failure, the test will be made as a part of quality conformance inspection, part 2, code level D, with an AQL of 6.5. The regular "12-calendar month" double sampling plan shall be reinstated after three consecutive samples have been accepted.

10. Revision letters are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - ER
Navy - EC
Air Force - 11, 17, 85, 99
DLA - ES

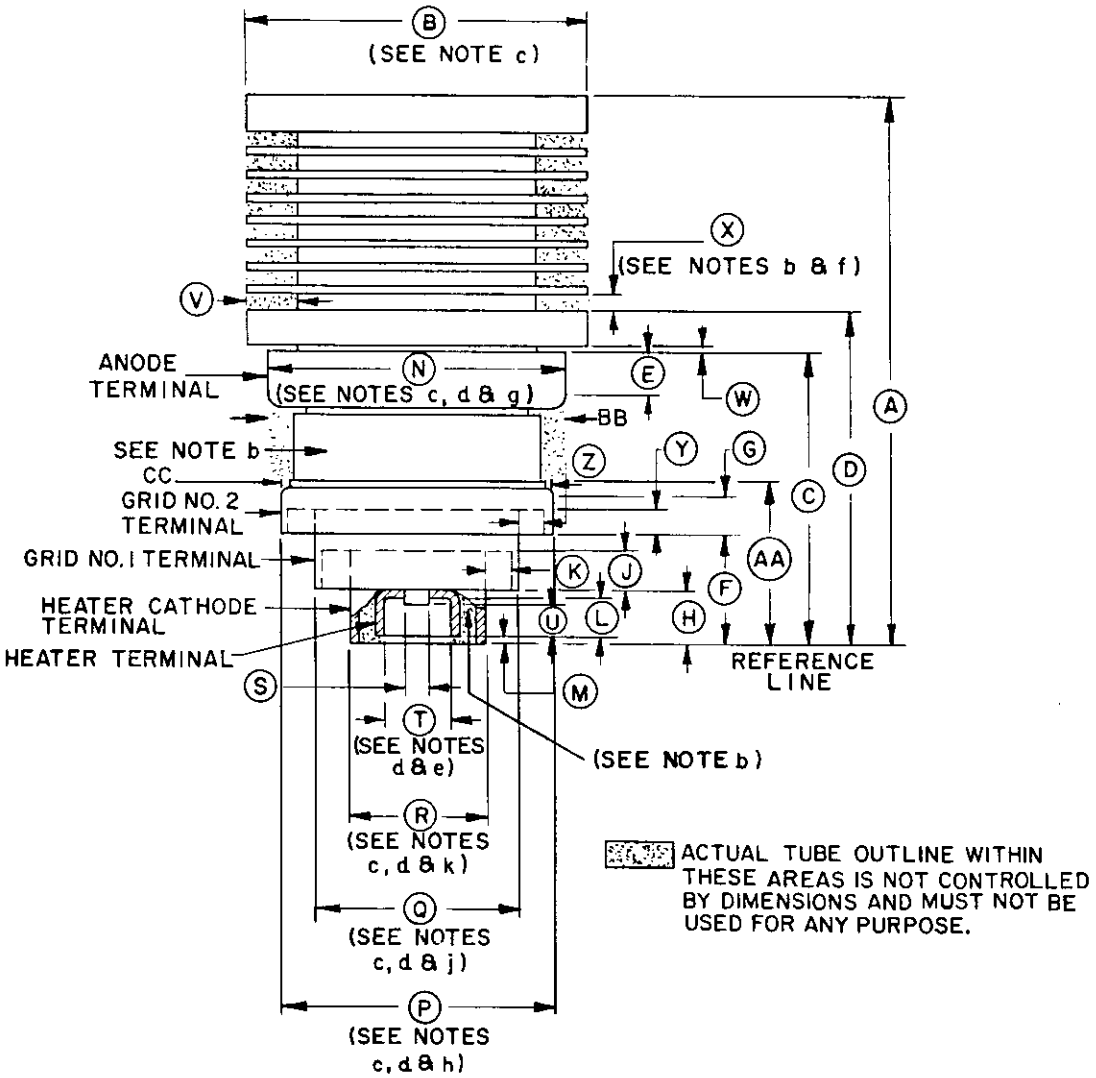
User activities:
Army - AR
Navy - AS, OS, MC, CG
Air Force - 19

Preparing activity:
Army - ER

Agent:
DLA - ES

(Project 5960-3257)

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G1: Grid 1 terminal (adjacent to cathode and heater terminal).
G2: Grid 2 terminal (adjacent to anode contact surface).
P: Anode-terminal contact surface (adjacent to radiator).
H: Heater terminals (cup at cathode end and cathode terminal).
K: Cathode terminal (end opposite radiator).

FIGURE 1. Outline drawing of electron tube types 6816 and 6884.

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses (see note a)		Notes
	Minimum	Maximum	
Quality conformace inspection, part 1			d, q
N	1.085 (27.56)		
P	.985 (25.02)		
Q	.735 (18.67)		
R	.480 (12.19)		
T	.240 (6.10)	.260 (6.60)	
Quality conformance inspection, part 2			
A	1.830 (46.48)	1.930 (49.02)	
B	1.235 (31.37)	1.265 (32.13)	
Quality conformance inspection, part 3 (see note r)			
C	1.000 (25.40)	1.060 (26.92)	
D	1.090 (27.69)	1.180 (29.97)	
E	.165 (4.19)		
F	.350 (8.89)	.390 (9.91)	
G	.140 (3.56)		
H	.160 (4.06)	.190 (4.83)	
J	.120 (3.05)		
K	.095 (2.41)		
L	.100 (2.54)		
M		.050 (1.27)	
S		.072 (1.83)	
U	.054 (1.37)		
V	.200 (5.08)		
W	.035 (.89)		
X	.050 (1.27)		
Y	.060 (1.52)		
Z	.090 (2.29)		
AA	.600 (15.24)		
BB			o
CC			p

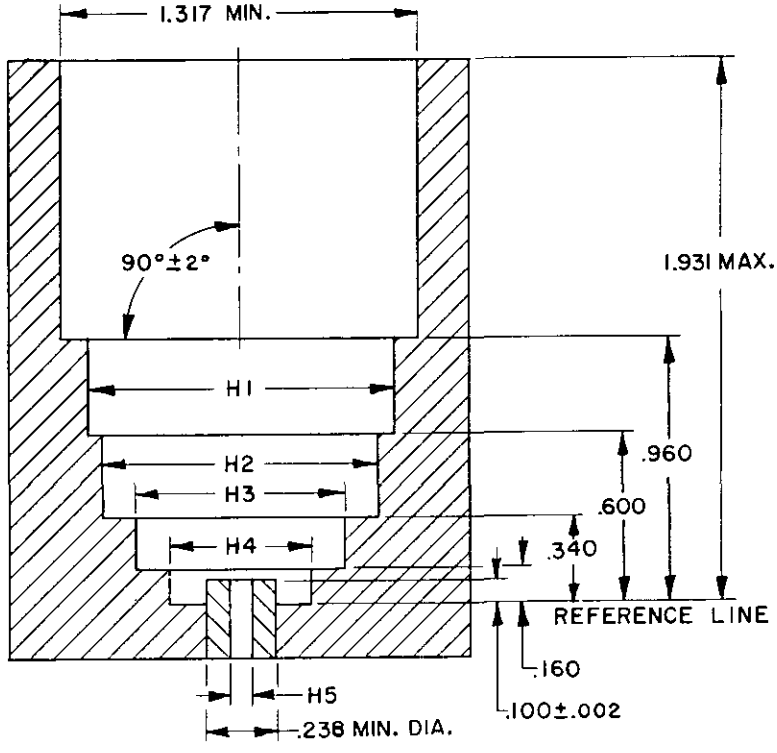
FIGURE 1. Outline drawing of electron tube types 6816 and 6884 - Continued.

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NOTES:
a. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
b. Areas between cooling fins, area between anode terminal and grid 2 terminal, and area between heater terminal and heater-cathode terminal shall not be used for any purpose.
c. Maximum dimensions shall be determined by gage No. 4 (see figure 2).
d. With cylindrical surfaces of anode terminal, screen grid terminal, control grid terminal, heater cathode terminal, and heater terminal clean, smooth, and free from burrs, the tube shall enter gage No. 4 (see figure 2). Axes of cylindrical holes H1 and H5 and axis of post P shall be concentric within .001 (.03 mm). Seating shall be determined by failure of gage .125 (3.18 mm) wide and .010 (.25 mm) thick to enter between heater cathode terminal and bottom surface of hole H4. A slot in the gage is provided to permit this measurement to be made.
e. Minimum dimension shall be determined by gage No. 4 (see figure 2).
f. This dimension shall be inspected by inserting gage .050 (1.27 mm) thick minimum between the two fins.
g. Dimension "N" applies to dimension "E" only.
h. Dimension "P" applies to dimension "G" only.
j. Dimension "Q" applies to dimension "F" minus dimension "H" only.
k. Dimension "R" applies to dimension "H" only.
m. In area included between anode terminal and grid 2 terminal, no part of tube shall exceed dimension "P".
o. On any one tube this dimension shall never be greater than "N".
p. On any one tube this dimension shall never be greater than "P".
q. The AQL for all dimensions listed under quality conformance inspection, part 1, shall be 1.0., inspection level I.
r. Dimensions shall be checked during the initial production and once each succeeding 12-calendar months in which there is production. A regular double sampling plan shall be used, with the first three tubes with an acceptance number of zero, and a second sample of three tubes with a combined acceptance number of one. In the event of failure, the test will be made as a part of quality conformance inspection, part 2, code level D, with an AQL of 6.5. The regular "12-calendar month" double sampling plan shall be reinstated after three consecutive samples have been accepted.

FIGURE 1. Outline drawing of electron tube types 6816 and 6884 - Continued.

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Hole	Limits	
	Min	Max
H1	--	1.120
H2	--	1.020
H3	--	.765
H4	--	.520
H5	--	.072

FIGURE 2. Gage No. 4.

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